TECH	7.3 How can materials and processes be used to make commercial products?	
	a. Demonstrate an understanding of the industrial processes and machinery used for manufacturing component parts in various materials, including: i. polymer moulding methods, such as injection moulding, blow moulding, compression moulding and thermoforming ii. metal casting methods such as sand casting and die casting iii. sheet metal forming methods using equipment such as punches, rollers, shears and stamping machines.	
	b. Demonstrate an understanding of the industrial methods used for assembling electronic products, including: i. surface mount technology (SMT) ii. PCB assembly using solder stencils, pick-and-place machines and reflow soldering ovens.	

	Considerations	Maths & Science
	c. Demonstrate an understanding of the benefits and flexibility of using computer-controlled machinery during industrial production, such as: • automated material handling systems • robot arms to stack, assemble, join and paint parts.	
	d. Understand the necessity for manufacturers to optimise the use of materials and production processes, such as: • economical cutting and costing, ensuring cost effective production for viability • working to a budget through efficient manufacture and making the best use of labour and capital throughout the design and manufacturing process.